M5B-E3 / ONE INTEGRATED 144MHz TRUE-AM EVALUATION PCB / NOT RELEASED
PHA-1+ and PD54008L-E retained; documented ST output match and elliptic LPF.
2S9Vboost OR4S9V/7.5Vbuck exclusive populations. Separate3.3Vtraining.
Driver: independent5.456V/5V regulation and18V-rated switch. Power-only qualification beforeR952/MMIC.
Input match has bounded same-board tuning;RF gain/AM/thermal/endurance tests NOT_RUN.
Review-only PCB/Gerbers. No ordering,fabrication,assembly or RF operation authorized.

01 Interface and local SI5351 source

File: 01_interface_source.kicad_sch

02 Minimal independent fail-off and captured ARM context

File: 02_interlocks.kicad_sch

02 Minimal independent fail-off and captured ARM context continued

File: 02_interlocksb1.kicad_sch

03 External protected pack and exclusive conversion populations

File: 03_input_power.kicad_sch

03 External protected pack and exclusive conversion populations continued

File: 03_input_powerb1.kicad_sch

04 2S boost and regulated MMIC rail

File: 04_boost_driver_supply.kicad_sch

05 Sine reconstruction and series AM modulator

File: 05_am.kicad_sch

05 Sine reconstruction and series AM modulator continued

File: 05_amb1.kicad_sch

06 Source filter, economical driver and fixed bias

File: 06_source_driver_bias.kicad_sch

06 Source filter, economical driver and fixed bias continued

File: 06_source_driver_biasb1.kicad_sch

07 Independent very-low-power sine-AM training path

File: 07_training.kicad_sch

07 Independent very-low-power sine-AM training path continued

File: 07_trainingb1.kicad_sch

08 Local voltage, current and thermal trip

File: 08_protection.kicad_sch

08 Local voltage, current and thermal trip continued

File: 08_protectionb1.kicad_sch

06b ST final-stage matching derivative

File: 06b_st_matching.kicad_sch

06c Original ST elliptic harmonic LPF

File: 06c_st_lpf.kicad_sch

10 Local physical switches and clock sequencing

File: 10_enable_detail.kicad_sch

11 External supplies and integrated commissioning

File: 11_external_contract.kicad_sch

12 Independent driver regulator and integrated alignment links

File: 12_m5b_qualification.kicad_sch

ARDF-TX / NOT RELEASED

Sheet: /
File: RF2M.kicad_sch

Title: ARDF-TX 2 m A2A evaluation schematic

Size: A2	Date: 2026-10-17	**Rev: M5B-E3**
KiCad E.D.A. 10.0.6		Id: 1/20

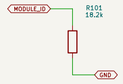
01 Interface and local Si5351 source

M5B-E3 / Integrated evaluation PCB / NOT RELEASED / hardware tests NOT_RUN

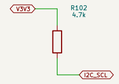
J101
Controller 20 pin



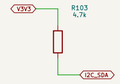
R101
18.2k



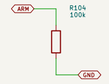
R102
4.7k



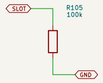
R103
4.7k



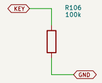
R104
100k



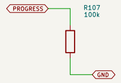
R105
100k



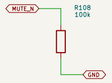
R106
100k



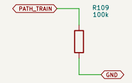
R107
100k



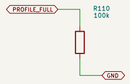
R108
100k



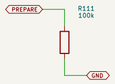
R109
100k



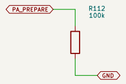
R110
100k



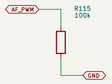
R111
100k



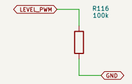
R112
100k



R115
100k



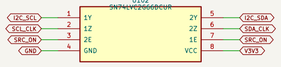
R116
100k



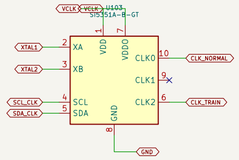
U101
TPS22919DCNR



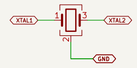
U102
SN74LVC2G66DCUR



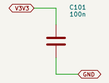
U103
SI5351A-B-GT



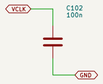
Y101
25MHz CL10p 10/10ppm



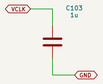
C101
100n



C102
100n



C103
1u



TPA01
CLK_NORMAL



TPA02
CLK_TRAIN



M5B-E3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

ARDF-TX

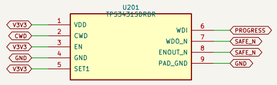
Sheet: /01 Interface and local Si5351 source/
File: 01_Interface_source.kicad_sch

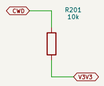
Title: 01 Interface and local Si5351 source

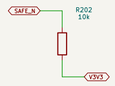
Size: A2	Date: 2026-10-17	**Rev: M5B-E3**
KiCad E.D.A. 10.0.6		Id: 2/20

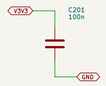
02 Minimal independent fail-off and captured ARM context

M5B-E3 / integrated evaluation PCB / NOT RELEASED / hardware tests NOT_RUN

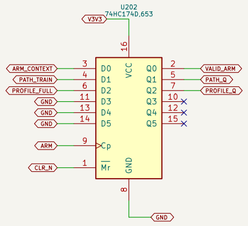














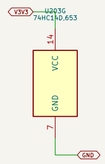


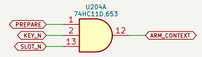




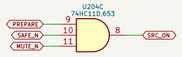


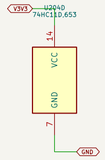


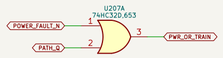
















M5B-E3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

ARDF-TX

Sheet: /02 Minimal independent fail-off and captured ARM context/
File: 02_Interlocks.kicad_sch

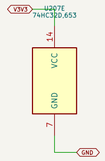
Title: 02 Minimal independent fail-off and captured ARM context

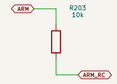
Size: A2	Date: 2026-10-17	**Rev: M5B-E3**
KiCad E.D.A. 10.0.6		Id: 3/20

02 Minimal independent fail-off and captured ARM context continued

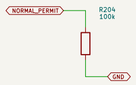
M5B-E3 / integrated evaluation PCB / NOT RELEASED / hardware tests NOT_RUN

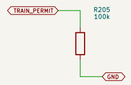




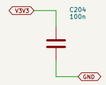


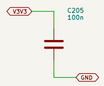


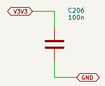


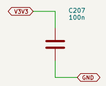


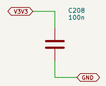


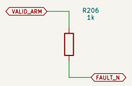


























M5B-E3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

ARDF-TX

Sheet: /02 Minimal independent fail-off and captured ARM context continued/
File: 02_Interlocksb1.kicad_sch

Title: 02 Minimal independent fail-off and captured ARM context continued

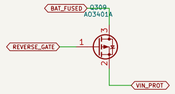
Size: A2	Date: 2026-10-17	**Rev: M5B-E3**
KiCad E.D.A. 10.0.6		Id: 4/20

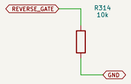
03 External protected pack and exclusive conversion populations

M5B-E3 / integrated evaluation PCB / NOT RELEASED / hardware tests NOT_RUN



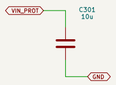


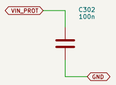


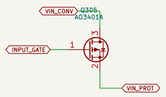


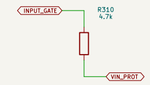


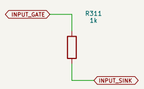


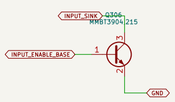


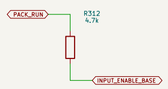


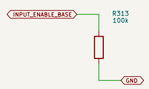






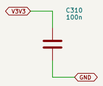




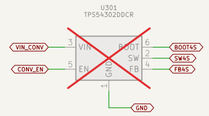




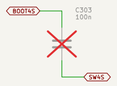






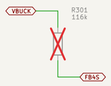




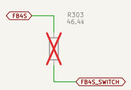


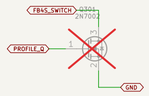


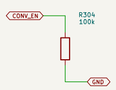


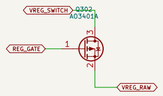


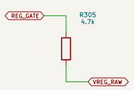












M5B-E3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

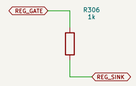
ARDF-TX

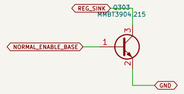
Sheet: /03 External protected pack and exclusive conversion populations/
File: 03_Input_power.kicad_sch

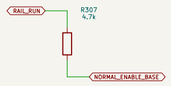
Title: 03 External protected pack and exclusive conversion populations

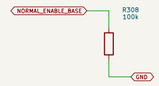
Size: A2	Date: 2026-10-17	**Rev: M5B-E3**
KiCad E.D.A. 10.0.6		Id: 5/20

M5B-E3 / integrated evaluation PCB / NOT RELEASED / hardware tests NOT_RUN

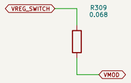






















M5B-E3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

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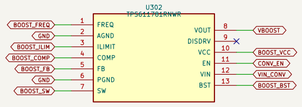
Sheet: /03 External protected pack and exclusive conversion populations continued/
File: 03_input_powerb1.kicad_sch

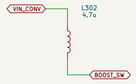
Title: 03 External protected pack and exclusive conversion populations continued

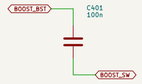
Size: A2	Date: 2026-10-17	**Rev: M5B-E3**
KiCad E.D.A. 10.0.6		Id: 6/20

04 2S boost and regulated MMIC rail

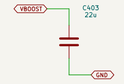
M5B-E3 / integrated evaluation PCB / NOT RELEASED / hardware tests NOT_RUN



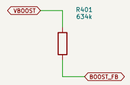


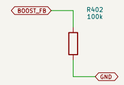


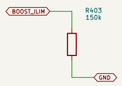


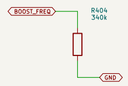


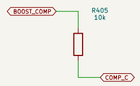


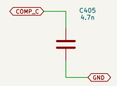


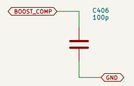


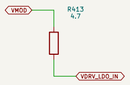


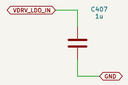




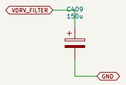
















M5B-E3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

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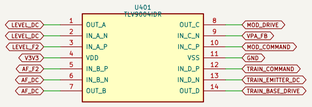
Sheet: /04 2S boost and regulated MMIC rail/
File: 04_boost_driver_supply.kicad_sch

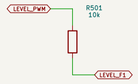
Title: 04 2S boost and regulated MMIC rail

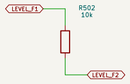
Size: A2	Date: 2026-10-17	**Rev: M5B-E3**
KiCad E.D.A. 10.0.6		Id: 7/20

05 Sine reconstruction and series AM modulator

M5B-E3 / integrated evaluation PCB / NOT RELEASED / hardware tests NOT_RUN

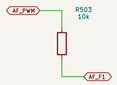


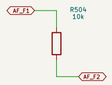






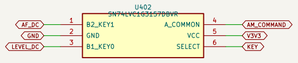


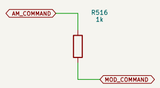


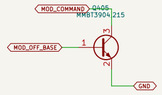


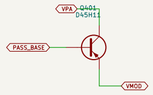


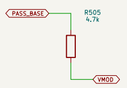


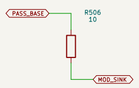


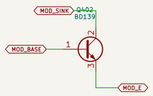


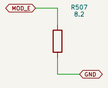


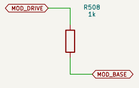


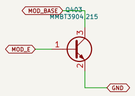


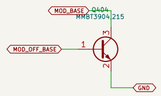


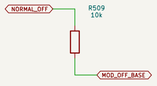


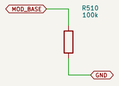


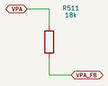


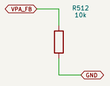


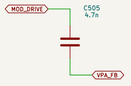




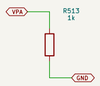




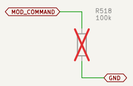


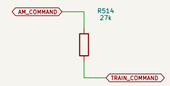












M5B-E3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

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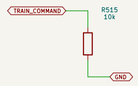
Sheet: /05 Sine reconstruction and series AM modulator/
File: 05_am.kicad_sch

Title: 05 Sine reconstruction and series AM modulator

Size: A2	Date: 2026-10-17	**Rev: M5B-E3**
KiCad E.D.A. 10.0.6		Id: 8/20

05 Sine reconstruction and series AM modulator continued

M5B-E3 / integrated evaluation PCB / NOT RELEASED / hardware tests NOT_RUN













M5B-E3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

ARDF-TX

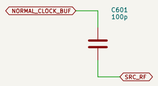
Sheet: /05 Sine reconstruction and series AM modulator continued/
File: 05_amb1.kicad_sch

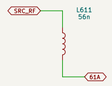
Title: 05 Sine reconstruction and series AM modulator continued

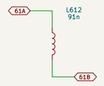
Size: A2	Date: 2026-10-17	**Rev: M5B-E3**
KiCad E.D.A. 10.0.6		Id: 9/20

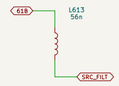
06 Source filter, economical driver and fixed bias

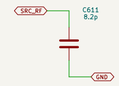
M5B-E3 / integrated evaluation PCB / NOT RELEASED / hardware tests NOT_RUN





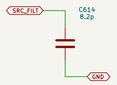


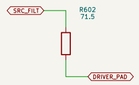


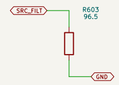


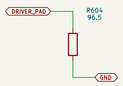


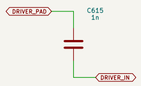


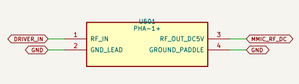


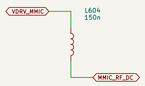


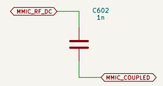


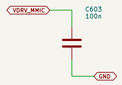


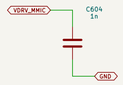


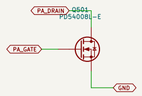


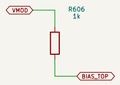


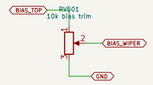




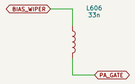


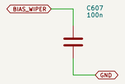


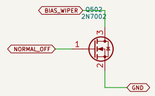














M5B-E3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

ARDF-TX

Sheet: /06 Source filter, economical driver and fixed bias/
File: 06_source_driver_bias.kicad_sch

Title: 06 Source filter, economical driver and fixed bias

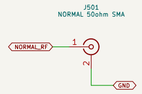
Size: A2	Date: 2026-10-17	**Rev: M5B-E3**
KiCad E.D.A. 10.0.6		Id: 10/20

06 Source filter, economical driver and fixed bias continued

M5B-E3 / integrated evaluation PCB / NOT RELEASED / hardware tests NOT_RUN









M5B-E3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

ARDF-TX

Sheet: /06 Source filter, economical driver and fixed bias continued/
File: 06_source_driver_biasb1.kicad_sch

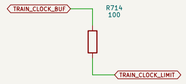
Title: 06 Source filter, economical driver and fixed bias continued

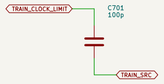
Size: A2	Date: 2026-10-17	**Rev: M5B-E3**
KiCad E.D.A. 10.0.6		Id: 11/20

07 Independent very-low-power sine-AM training path

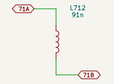
M5B-E3 / integrated evaluation PCB / NOT RELEASED / hardware tests NOT_RUN

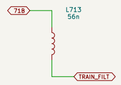


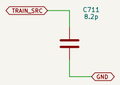


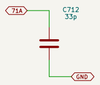


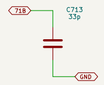




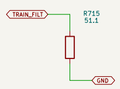


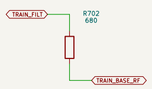


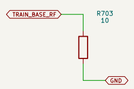


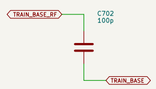


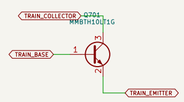


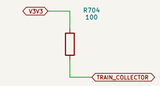


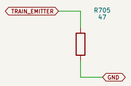


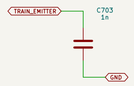


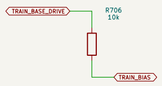


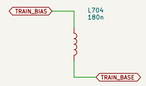


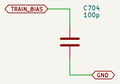


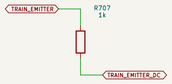


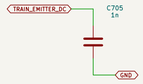


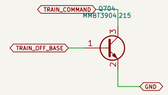


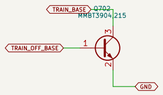


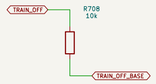


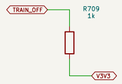


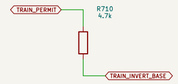


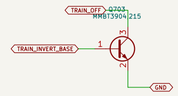


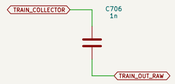












M5B-E3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

ARDF-TX

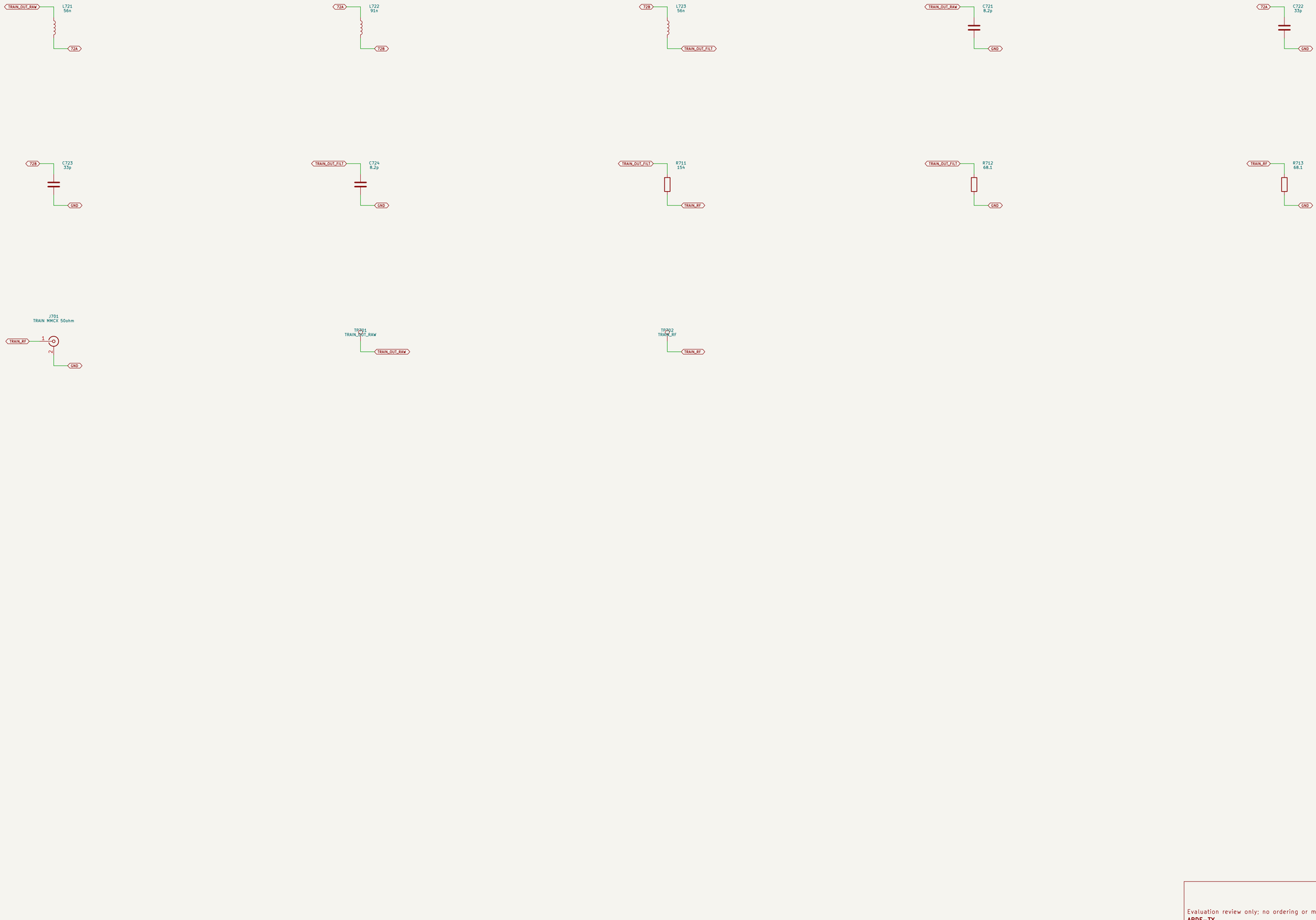
Sheet: /07 Independent very-low-power sine-AM training path/
File: 07_training.kicad_sch

Title: 07 Independent very-low-power sine-AM training path

Size: A2	Date: 2026-10-17	**Rev: M5B-E3**
KiCad E.D.A. 10.0.6		Id: 12/20

07 Independent very-low-power sine-AM training path continued

M5B-E3 / integrated evaluation PCB / NOT RELEASED / hardware tests NOT_RUN



M5B-E3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX

Sheet: /07 Independent very-low-power sine-AM training path continued/
File: 07_trainingb1.kicad_sch

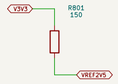
Title: 07 Independent very-low-power sine-AM training path continued

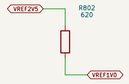
Size: A2	Date: 2026-10-17	**Rev: M5B-E3**
KiCad E.D.A. 10.0.6		Id: 13/20

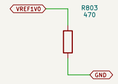
08 Local voltage, current and thermal trip

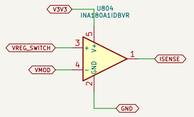
M5B-E3 / integrated evaluation PCB / NOT RELEASED / hardware tests NOT_RUN

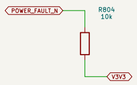


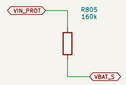


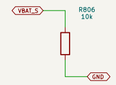


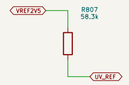


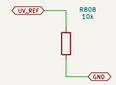


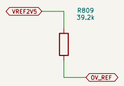


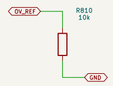


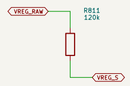


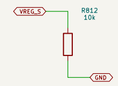


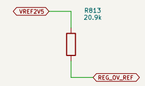


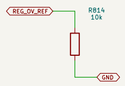


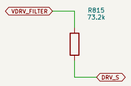


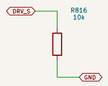


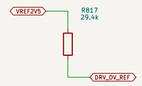


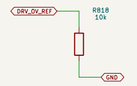


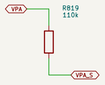


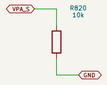


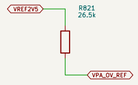


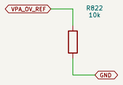


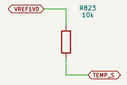


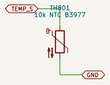


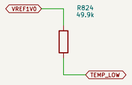


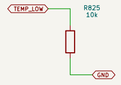


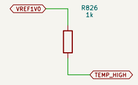


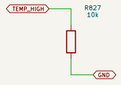












M5B-E3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

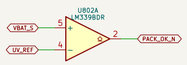
ARDF-TX

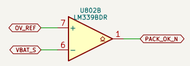
Sheet: /08 Local voltage, current and thermal trip/
File: 08_protection.kicad_sch

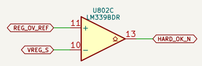
Title: 08 Local voltage, current and thermal trip

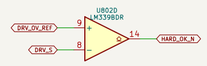
Size: A2	Date: 2026-10-17	**Rev: M5B-E3**
KiCad E.D.A. 10.0.6		Id: 14/20

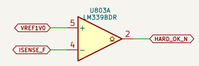
M5B-E3 / integrated evaluation PCB / NOT RELEASED / hardware tests NOT_RUN

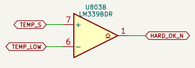


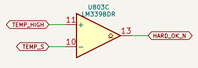


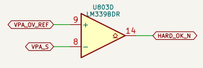


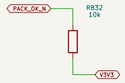


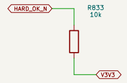










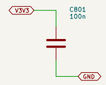


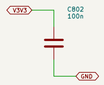


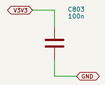


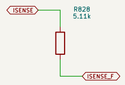


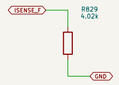


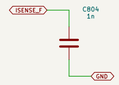






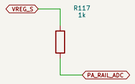


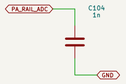














M5B-E3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

ARDF-TX

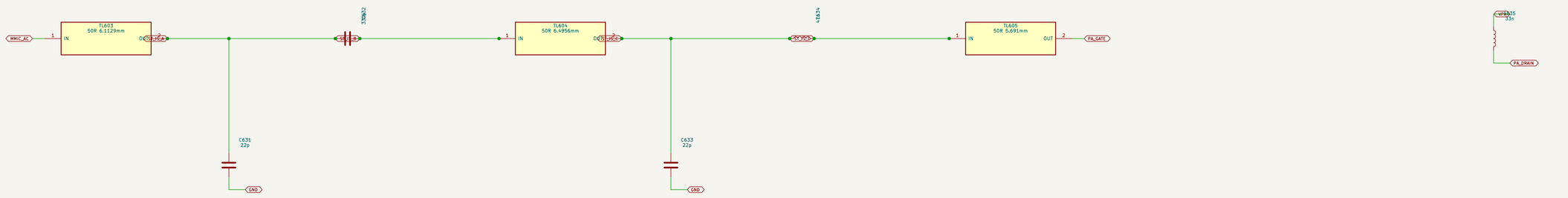
Sheet: /08 Local voltage, current and thermal trip continued/
File: 08_protectionb1.kicad_sch

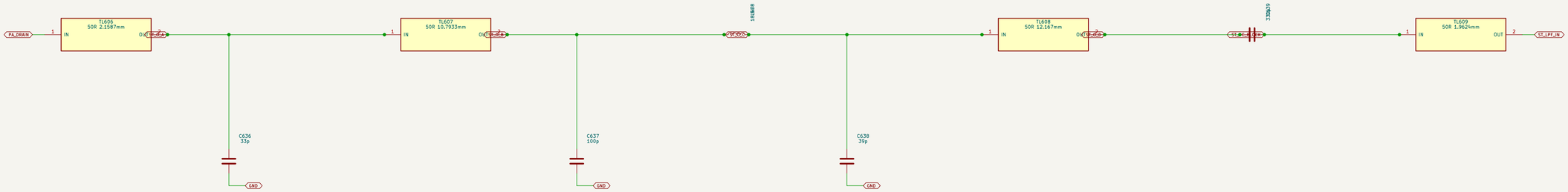
Title: 08 Local voltage, current and thermal trip continued

Size: A2	Date: 2026-10-17	**Rev: M5B-E3**
KiCad E.D.A. 10.0.6		Id: 15/20

06b ST final-stage matching derivative

ST DocID17343 Rev1 Figs16/17/14. Traced topology; changed MMIC drive and supplyAM remain UNQUALIFIED.
TLine symbols are PCB microstrip lengths, NOT purchased components or zero-length wire. No hardware measurements exist.





Input matching: original interstage source was PD84001 drain, not a50ohm MMIC. No claim of unchanged match or sufficient gain.
Output matching/LPF retained. L633.3s drain feed is a disclosed substitution for the contradictory35.5n/22n ST BOM entry.
Use one integrated end-to-end commissioning procedure after independent electrical/layout review. No separate PA coupons.

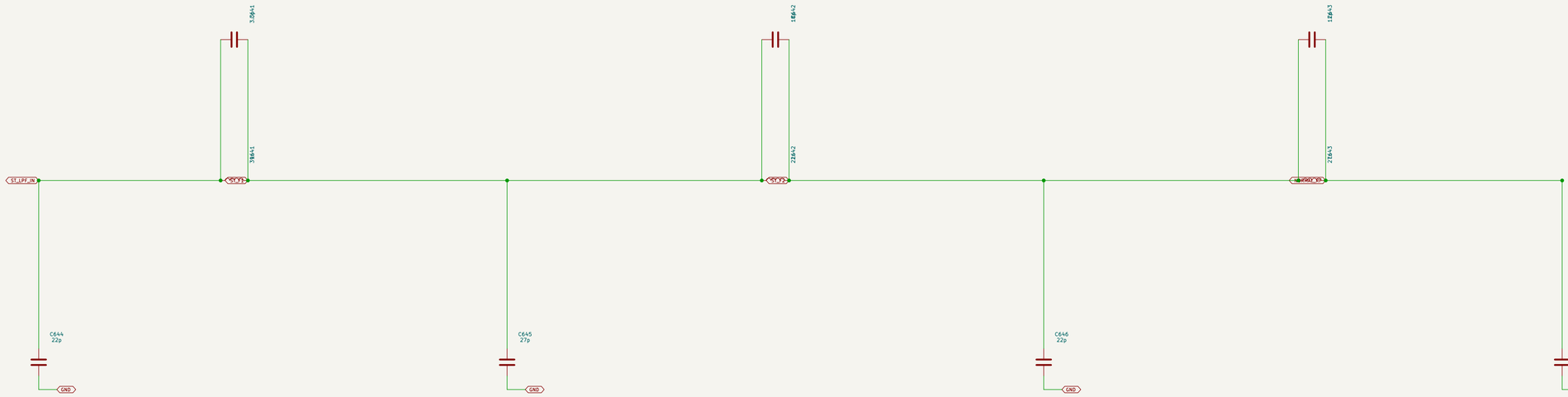
M5B-E3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX	
Sheet: /06b ST final-stage matching derivative/	
File: 06b_st_matching.kicad_sch	
Title: 06b ST final-stage matching derivative	
Size: A2	Date: 2026-10-17
KiCad E.D.A. 10.0.6	**Rev: M5B-E3**
	Id: 16/20

06c Original ST elliptic harmonic LPF

ST DocID17343 Rev1 Figs16/17/14. Traced topology; changed MMIC drive and supplyAM remain UNQUALIFIED.
TLine symbols are PCB microstrip lengths, NOT purchased components or zero-length wire. No hardware measurements exist.



Input matching: original interstage source was PD84001 drain, not a50ohm MMIC. No claim of unchanged match or sufficient gain.
Output matching/LPF retained. L63533s drain feed is a disclosed substitution for the contradictory35.5n/22n ST BOM entry.
Use one integrated end-to-end commissioning procedure after independent electrical/layout review. No separate PA coupons.

M5B-E3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX

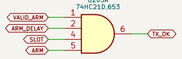
Sheet: /06c Original ST elliptic harmonic LPF/
File: 06c_st_lpf.kicad_sch

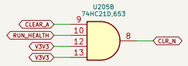
Title: 06c Original ST elliptic harmonic LPF

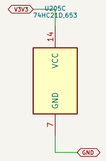
Size: A2	Date: 2026-10-17	**Rev: M5B-E3**
KiCad E.D.A. 10.0.6		Id: 17/20

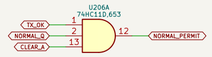
10 Local physical switches and clock sequencing

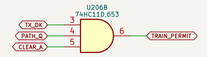
M5B-E3 / integrated evaluation PCB / NOT RELEASED / hardware tests NOT_RUN



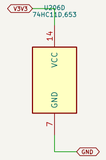




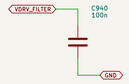


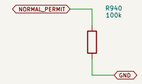


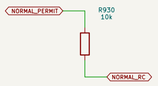


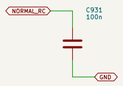


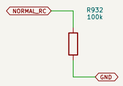




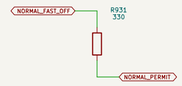


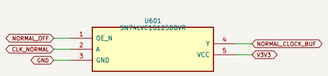


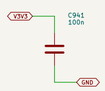














M5B-E3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX

Sheet: /10 Local physical switches and clock sequencing/
File: 10_enable_detail.kicad_sch

Title: 10 Local physical switches and clock sequencing

Size: A2	Date: 2026-10-17	**Rev: M5B-E3**
KiCad E.D.A. 10.0.6		Id: 18/20

11 External supplies and integrated commissioning

M5B-E3 / integrated evaluation PCB / NOT RELEASED / hardware tests NOT_RUN





















M5B-E3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

ARDF-TX

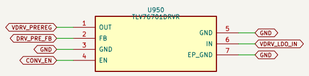
Sheet: /11 External supplies and integrated commissioning/
File: 11_external_contract.kicad_sch

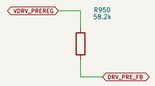
Title: 11 External supplies and integrated commissioning

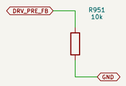
Size: A2	Date: 2026-10-17	**Rev: M5B-E3**
KiCad E.D.A. 10.0.6		Id: 19/20

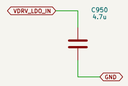
12 Independent driver regulator and integrated alignment links

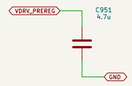
Pre-regulation5.456V then existing5V LDO. Rated18V switch. No20us RC assumption is claimed as protection proof.
Fit driver link only after power-only qualification. RF link and optional50ohm load permit alignment on THIS board.

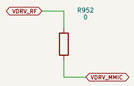


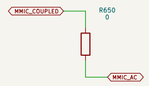


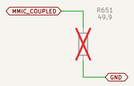






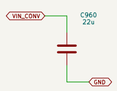


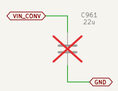












M5B-E3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

ARDF-TX

Sheet: /12 Independent driver regulator and integrated alignment links/
File: 12_m5b_qualification.kicad_sch

Title: 12 Independent driver regulator and integrated alignment links

Size: A2	Date: 2026-10-17	**Rev: M5B-E3**
KiCad E.D.A. 10.0.6		Id: 20/20